Quantification: 548_hyp_HP_z14c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	17
interior lacuna count	12
frame-edge lacuna count	5

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2845.42	2798.5	1991.25	px^2
major_axis_length_px	lacuna shape	83.0974	87.8292	34.3419	px
minor_axis_length_px	lacuna shape	46.5755	50.3226	19.1752	px
aspect_ratio	lacuna shape	1.826	1.9787	0.4126	unitless
eccentricity	lacuna shape	0.7997	0.8629	0.1126	unitless
solidity	lacuna shape	0.7785	0.7974	0.0877	unitless
roots_count	headline	10	9	5.0272	count
ring_length_r30_px	headline	337.5	304.5	130.494	px
ring_length_r60_px	reported	694.25	642.5	220.211	px
owned_length_px	ownership-dependent	1161.93	1072.01	776.702	px
edge_count	ownership-dependent	58.3333	50.5	38.3224	count
mean_edge_length_px	ownership-dependent	19.7535	19.0921	5.0814	px
perimeter_px	normalised	291.18	308.898	147.809	px
ring_area_r30_px2	normalised	9715.83	9319.5	3389.83	px^2
ring_area_r60_px2	normalised	21682.5	21970.5	6642.82	px^2
in_frame_fraction_r30	normalised	0.9735	1	0.0598	unitless
in_frame_fraction_r60	normalised	0.9455	1	0.106	unitless
ring_density_r30	normalised	0.0349598	0.034731	0.00576815	px^-1
ring_density_r60	normalised	0.0324104	0.0326799	0.00479453	px^-1
roots_per_100px_perimeter	normalised	3.7423	3.4684	1.4865	per 100 px
ring_attached_length_r30_px	network v2	298.75	267.5	142.564	px
ring_attached_length_r60_px	network v2	540.917	481.5	228.61	px
ring_length_w_r30_px	network v2	386.602	352.428	148.145	px
ring_length_w_r60_px	network v2	792.589	745.11	249.351	px
sholl_crossings_r10	network v2	7.9167	8.5	3.1176	count
sholl_crossings_r20	network v2	9.5833	8.5	3.7528	count
sholl_crossings_r30	network v2	9	8.5	4.1121	count
loop_count	loops, report only	4.0833	3	3.6045	count
loop_edge_share	loops, report only	0.2193	0.2424	0.1159	share
ring_width_mean_r30_px	width	7.6994	7.5089	0.5385	px

Per field
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measure	value	units
analysed_area_px2	1.0013e+06	px^2
total_skeleton_length_px	27827	px
canalicular_length_density_per_px	0.0277908	px^-1
junction_count	588	count
junction_density_per_px2	0.00058724	px^-2
lacuna_count	17	count

lacunae_per_px2	1.698e-05	px^-2
skeleton_component_count	386	count
mean_component_length_px	72.0907	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0316294	px^-1
field_density_without_flagged_per_px	0.0277908	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	248	px
unattached_component_count	14	count
canalicular_length_density_attached_per_px	0.0277908	px^-1
field_length_density_with_unattached_per_px	0.0280385	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	76	count
loop_edge_share	0.1984	share
edge_count	1532	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8411	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	27827	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2147	91.4001	39.1166	2.3366	0.9038	0.7373	0.5027	31.0005	542	True
2	2407	118.141	31.4268	3.7592	0.964	0.7314	0.0789	45.1836	688.173	True
3	3074	97.1472	47.5979	2.041	0.8717	0.7951	0.7084	40.3224	458.502	False
4	4688	114.763	56.7615	2.0218	0.8691	0.8409	-0.5667	73.3957	882.268	False
5	2523	87.2464	53.9799	1.6163	0.7856	0.696	0.2148	200.3	119.131	False
6	885	49.967	25.4783	1.9612	0.8602	0.8157	0.1758	321.858	27.7831	False
7	865	60.9988	23.7735	2.5658	0.9209	0.67	-0.6206	306.805	864.442	False
8	5590	100.499	74.9477	1.3409	0.6662	0.8931	0.3458	390.83	706.214	False
9	3806	88.412	63.5037	1.3922	0.6958	0.7996	0.5855	456.034	447.32	False
10	1368	68.8013	32.6693	2.106	0.8801	0.7621	0.1327	437.326	13.0395	True
11	818	48.7035	23.1908	2.1001	0.8794	0.8503	0.5594	490.792	650.099	False
12	583	30.9882	25.7454	1.2036	0.5565	0.8807	-0.6814	486.81	163.602	False
13	6173	146.612	73.4479	1.9961	0.8655	0.639	0.4437	581.134	750.702	False
14	1268	55.1715	37.4316	1.4739	0.7346	0.792	0.9741	664.745	233.587	False
15	1408	71.6194	28.9459	2.4743	0.9147	0.8167	1.0982	723.629	987.69	True
16	3872	116.659	53.0473	2.1992	0.8906	0.6702	-1.541	793.571	878.724	False
17	5800	180.953	48.7781	3.7097	0.963	0.7753	-1.1774	895.486	957.446	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

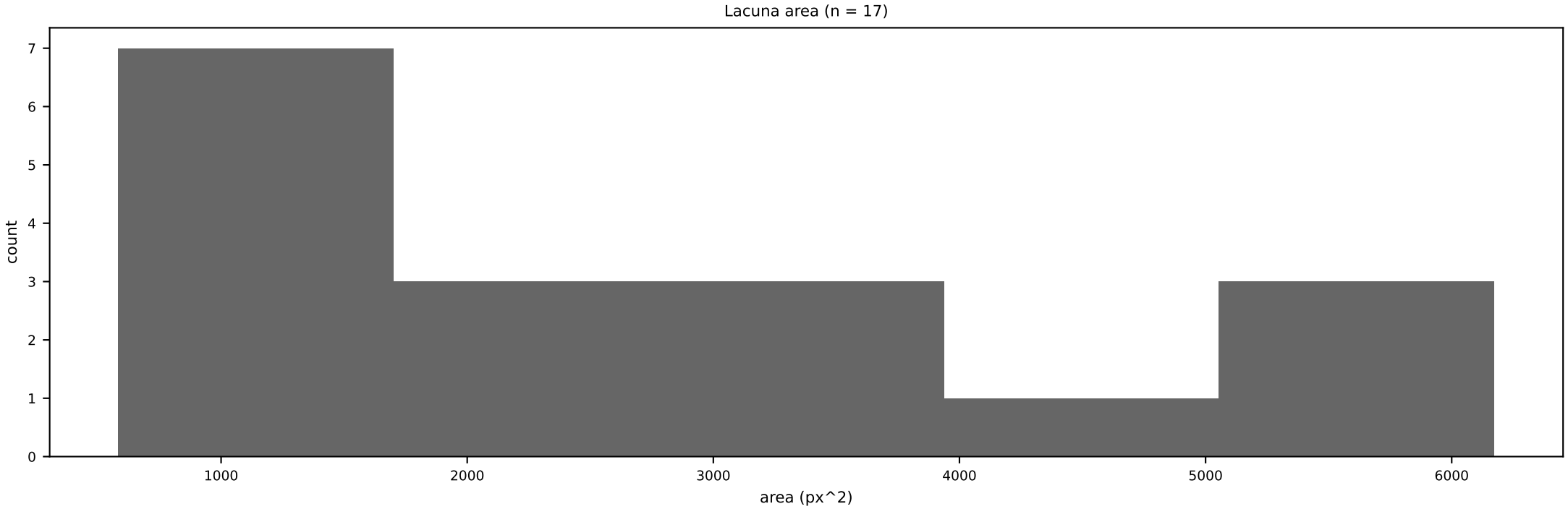
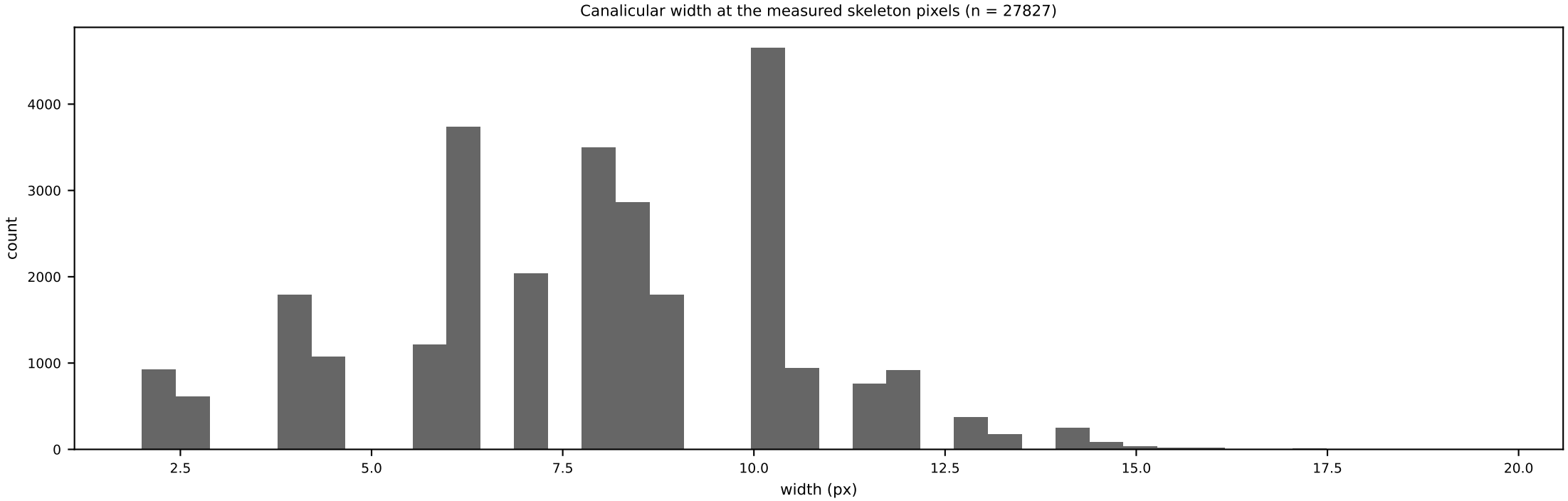
1	7	191	343	781.281	31	25.2026	6.9981
2	10	354	744	880.637	44	20.0145	7.8088
3	10	241	509	1773.19	91	19.4856	7.7676
4	7	288	596	884.205	27	32.7483	8.7175
5	9	328	766	1171.69	49	23.9121	7.4401
6	6	244	452	105.012	8	13.1265	7.2348
7	12	321	689	1428.11	80	17.8514	7.3687
8	9	469	961	2883.77	138	20.8969	8.7134
9	10	353	836	972.323	52	18.6985	7.1274
10	2	98	214	367.463	18	20.4146	7.3659
11	4	218	399	463.345	22	21.0611	7.4299
12	7	184	566	283.451	21	13.4977	7.9669
13	22	607	1128	1944.31	98	19.8399	7.5777
14	7	274	570	813.784	44	18.4951	7.2242
15	8	311	574	1513.02	74	20.4463	7.755
16	17	523	859	1220.02	70	17.4288	7.8248
17	15	423	787	714.156	44	16.2308	8.2057

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	314.877	6967	12578	0.7247	0.6734	0.027415	0.0272698	2.2231
2	330.693	8703	19577	0.7916	0.7301	0.0406756	0.0380038	3.0239
3	289.563	8243	15900	0.836	0.748	0.0292369	0.0320126	3.4535
4	351.161	12196	26656	0.9903	0.8855	0.0236143	0.0223589	1.9934
5	371.055	10396	25417	1	1	0.0315506	0.0301373	2.4255
6	148.61	5799	11539	0.8558	0.7127	0.0420762	0.0391715	4.0374
7	184.652	7467	20230	1	1	0.0429891	0.0340583	6.4987
8	352.919	11920	25165	1	1	0.0393456	0.038188	2.5502
9	328.233	11389	28087	1	1	0.0309948	0.0297647	3.0466
10	193.539	5102	10952	0.6366	0.5949	0.0192082	0.0195398	1.0334
11	137.196	6367	12630	1	1	0.0342391	0.0315915	2.9155
12	104.569	5596	16799	1	1	0.0328806	0.0336925	6.6942
13	631.595	16663	33826	1	1	0.036428	0.0333471	3.4832
14	199.865	7779	21047	1	1	0.035223	0.0270822	3.5024
15	197.48	6605	13178	0.8037	0.7245	0.0470855	0.0435574	4.051
16	394.747	12775	22894	1	1	0.0409393	0.0375207	4.3066
17	482.86	12649	23651	0.8273	0.7718	0.0334414	0.0332756	3.1065

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	181	317	208.64	386.563	4	3	5
2	256	369	410.333	863.637	4	8	11
3	188	413	275.451	575.931	5	7	4
4	278	483	336.363	692.382	9	9	10
5	290	575	368.492	855.34	10	9	9
6	187	292	274.137	510.375	8	8	5
7	291	580	378.99	797.838	6	10	6
8	469	926	537.345	1087.02	9	12	8
9	257	407	394.191	958.235	11	8	17
10	50	85	100.042	229.267	1	3	7
11	126	283	261.392	474.943	3	7	5
12	167	329	206.51	643.529	4	4	9
13	580	904	697.612	1291.34	11	16	14
14	255	480	318.907	643.872	6	8	7
15	306	518	353.25	648.63	5	8	9

16	497	819	589.831	980.264	13	17	14
17	367	619	480.789	886.423	12	10	12

548_hyp_HP_z14c1-2: distributions, pre-validation, px